

TEACHER COPY

Final P1 mock mark scheme

75 marks - full syllabus

YOUR AIM TODAY

Mark methods consistently and convert every lost mark into a correction task.

YOUR RULE

Keep closed until the timed mock is finished. Equivalent exact forms are acceptable.

Student: Rohan

Start time: _____ Finish time: _____

Date: _____

Mark allocation and thresholds

M=method, A=accuracy, B=independent result.

Section	Marks
Quadratics	7
Functions	7
Coordinate geometry	8
Circular measure	7
Trigonometry	15
Series	8
Differentiation	8
Integration/calculus	15
Total	75

WORKING TARGET

60+ secure; 52-59 strong with repairs; 42-51 developing; below 42 requires systematic reteaching before repeated full papers.

Quadratics and functions

Part	Answer and allocation	Marks
1(a)	$2(x-3)^2-7$. M1 factor 2; M1 square; A1.	3
1(b)	$x=3\pm\sqrt{14}/2$. M1 from square; A1 exact roots.	2
1(c)	$x<3-\sqrt{14}/2$ or $x>3+\sqrt{14}/2$. M1 roots/shape; A1.	2
2(a)	$fg=3x^2-11$. M1 order; A1.	2
2(b)	$gf=(3x+1)^2-4=9x^2+6x-3$. M1; A1.	2
2(c)	$g^{-1}(x)=\sqrt{x+4}$. M1; A1.	2
2(d)	Range $[-4,\infty)$. B1.	1

Coordinate and circular measure

Part	Answer and allocation	Marks
3(a)	$\frac{4}{3}$. M1+A1.	2
3(b)	10. M1+A1.	2
3(c)	$y-2=(\frac{4}{3})(x+1)$, e.g. $3y=4x+10$. M1+A1.	2
3(d)	Midpoint (2,6), perpendicular $m=-\frac{3}{4}$; $y-6=(-\frac{3}{4})(x-2)$. M1+A1.	2
4(a)	4π cm. M1+A1.	2
4(b)	12π cm ² . M1+A1.	2
4(c)	$12\pi-9\sqrt{3}$ cm ² . M1 triangle; M1 subtract; A1.	3

Trigonometry

Part	Answer and allocation	Marks
5(a)	$-\frac{1}{2}$ and -1 . B1 each.	2
5(b)	Amplitude 2; period $2\pi/3$. B1 each.	2
5(c)	$[-3,1]$. B1.	1
5(d)	$5\pi/6$, $7\pi/6$. B1 each.	2
6(a)	$1 - \cos^2 x = \sin^2 x$, then divide by $\sin x$. M1+A1.	2
6(b)	$(2\sin x - 1)(\sin x - 1) = 0$; $x = 30, 90, 150$ degrees. M1 factor; M1 cases; A1 solutions; A1 complete.	4
6(c)	$x = 22.5, 112.5$ degrees. M1+A1.	2

QUESTION 7

Series

Part	Answer and allocation	Marks
7(a)	$a+2d=11$, $a+7d=31$; $d=4, a=3$. M1 equations; A1 d; A1 a.	3
7(b)	$S_{20}=820$. M1 formula; A1.	2
7(c)	$C(6,3)2^3(-1)^3=-160$. M1 term; M1 powers/sign; A1.	3

Differentiation

Part	Answer and allocation	Marks
8(a)	$6x^2 - 18x + 12 = 6(x-1)(x-2)$. M1+A1.	2
8(b)	(1,5) and (2,4). M1 roots; A1 each coordinate pair.	3
8(c)	(1,5) maximum; (2,4) minimum. B1 each.	2
8(d)	$y=12x$. B1.	1

Integration and calculus

Part	Answer and allocation	Marks
9(a)	$2x^3 - 2x^2 + 3x + C$. M1+A1.	2
9(b)	$C=2$, so $y=2x^3 - 2x^2 + 3x + 2$. M1+A1.	2
9(c)	14. M1+A1.	2
9(d)	$V = \pi \int_0^4 x \, dx = 8\pi$. M1+A1.	2
10(a)	$x=0, 4$. B1.	1
10(b)	(2,4), maximum. M1 derivative/solve; A1 coordinate; A1 nature.	3
10(c)	$\int_0^4 (4x - x^2) \, dx = 32/3$. M1 limits; M1 primitive; A1.	3

AFTER MARKING

Enter topic percentages in the tracker. Redo every lost mark without looking, then assign a three-question retest to each topic below 80%.